

EXPERIMENT 8 – Deadlock Detection Algorithm

Aim: To implement the Deadlock Detection Algorithm.

C Program:

```
#include <stdio.h>

int main()
{
    int allocation[10][10], request[10][10];
    int available[10];
    int work[10];
    int finish[10];
    int n, m;
    int i, j, k;
    int found;

    printf("Enter Number of Processes: ");
    scanf("%d", &n);

    printf("Enter Number of Resource Types: ");
    scanf("%d", &m);

    printf("\nEnter Allocation Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &allocation[i][j]);

    printf("\nEnter Request Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &request[i][j]);

    printf("\nEnter Available Resources:\n");
    for(i = 0; i < m; i++)
    {
        scanf("%d", &available[i]);
        work[i] = available[i];
    }

    for(i = 0; i < n; i++) finish[i] = 0;

    do {
        found = 0;
        for(i = 0; i < n; i++)
        {
            if(finish[i] == 0)
            {
                for(j = 0; j < m; j++)
                    if(request[i][j] > work[j]) break;

                if(j == m)
                {
                    for(k = 0; k < m; k++)
                        work[k] += allocation[i][k];
                    finish[i] = 1;
                    found = 1;
                }
            }
        }
    } while(found);

    found = 0;
    printf("\nDeadlocked Processes:\n");
    for(i = 0; i < n; i++)
        if(finish[i] == 0) { printf("P%d ", i); found = 1; }

    if(found == 0) printf("No Deadlock Detected");
    printf("\n");

    return 0;
}
```

Compilation: `gcc deadlock.c -o deadlock`

Output (3 Processes, 2 Resources — deadlock scenario):

Deadlocked Processes:
P0 P1

Shell Script:

```
#!/bin/bash

echo "Enter number of processes:"
read n

echo "Enter deadlocked process numbers (if any):"
read processes

if [ -z "$processes" ]
then
    echo "No Deadlock Detected"
else
    echo "Deadlocked Processes:"
    echo "$processes"
fi
```

Result: Deadlock Detection Algorithm was implemented and verified successfully.
